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RETRACTABLE BUMPER ASSEMBLY WITH LOCK

Abstract

A vehicle-bumper assembly includes a bumper and a rod fixed relative to the bumper and elongated along an axis. A bracket is spaced from the bumper along the axis of the rod. The rod is movably engaged with the bracket **16** between a design position and a retracted position. A spring **18** is between the bumper and the bracket. The spring resiliently forces the bumper away from the bracket along the axis of the rod toward the design position. A keeper is releasably engaged with the bracket. The keeper maintains the rod in the design position when the keeper is engaged with the bracket. A lock is movable from a locked position in which the lock couples the keeper and the bracket and an unlocked position in which the lock is decoupled with the keeper and/or the bracket.

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Background/Summary

BACKGROUND

[0001] The Global Technology Regulation (GTR) and the New Car Assessment Program (NCAP) specify leg-injury criteria for pedestrian protection. The regulations are aimed at reducing the impact force to the legs of a pedestrian by a vehicle bumper during certain vehicle-pedestrian impacts.

[0002] Some vehicles, such as light duty trucks and sport utility vehicles (SUVs), for example, may have a bumper height that could lead to an uneven impact on the femur and/or tibia of the pedestrian by the vehicle bumper during certain vehicle-pedestrian impacts. For example, light duty trucks may have bumper heights to provide ground clearance to clear speed bumps, curbs, parking blocks, inclined driveway ramps, hills, rough roads, etc. Some vehicles with such bumper heights also have off-road capabilities that preclude having any components below the bumper. As such, a design for the vehicle front-end is investigated for pedestrian leg impact energy management while addressing ground clearance requirements.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0003] FIG. **1** is a perspective view of a portion of a vehicle including a vehicle-bumper assembly.

[0004] FIG. **2** is a perspective view of the vehicle with the vehicle-bumper assembly removed.

[0005] FIG. **3** is a partially exploded view of the vehicle-bumper assembly.

[0006] FIG. **4** is a top view of the vehicle-bumper assembly and a portion of a vehicle frame.

[0007] FIG. **5**A is a perspective view of a portion of the vehicle-bumper assembly with a lock in a locked position.

[0008] FIG. **5**B is a perspective with the lock in an unlocked position and a rod of the vehicle-bumper in a design position.

[0009] FIG. **5**C is a perspective view with the lock in the unlocked position and the rod in a retracted position.

[0010] FIG. **6** is a perspective view of another example of the vehicle-bumper assembly.

[0011] FIG. **7** is a block diagram of a system of the vehicle.

[0012] FIG. **8** is a flow chart of an example method.

DETAILED DESCRIPTION

[0013] A vehicle-bumper assembly includes a bumper and a rod fixed relative to the bumper and elongated along an axis. A bracket is spaced from the bumper along the axis of the rod. The rod is movably engaged with the bracket between a design position and a retracted position. A spring is between the bumper and the bracket. The spring resiliently forces the bumper away from the bracket along the axis of the rod toward the design position. A keeper is releasably engaged with the bracket. The keeper maintains the rod in the design position when the keeper is engaged with the bracket. A lock is movable between a locked position in which the lock couples the keeper and the bracket and an unlocked position in which the lock is decoupled from the keeper and/or the bracket.

[0014] The vehicle-bumper assembly may include a releasable connection between the keeper and bracket. The releasable connection may be designed to release the keeper from the bracket in response to certain frontal-vehicle impacts.

[0015] The vehicle-bumper assembly may include a shear pin between the keeper and the bracket.

[0016] The rod may extend through the bracket and the bracket is between the keeper and the bumper. The keeper may be disposed on the axis of the rod. The lock may include an actuator and a latch movable by the actuator. The latch may be movable by the actuator between the locked position in which the keeper is between the latch and the bracket and the unlocked position in which the latch is disengaged with the keeper. The actuator may be on the keeper.

[0017] The spring may be on the rod.

[0018] A vehicle may include a vehicle frame, a bumper, and a rod fixed relative to the bumper and elongated along an axis. A bracket is fixed relative to the vehicle frame and spaced from the bumper along the axis of the rod. The rod is movably engaged with the bracket between a design position and a retracted position. A spring is between the bumper and the bracket. The spring resiliently forces the bumper away from the bracket along the axis of the rod toward the design position. A keeper is releasably engaged with the bracket. The keeper maintains the rod in the design position when the keeper is engaged with the bracket. A lock is movable between a locked position in which the lock couples the keeper and the bracket and an unlocked position in which the lock decouples the keeper and/or the bracket.

[0019] The vehicle-bumper assembly may include a releasable connection between the keeper and bracket. The releasable connection may be designed to release the keeper from the bracket in response to certain frontal-vehicle impacts.

[0020] The vehicle-bumper assembly may include a shear pin between the keeper and the bracket.

[0021] The rod may extend through the bracket and the keeper is on a vehicle-rearward side of the bracket. The keeper may be disposed on the axis of the rod vehicle-rearward of the rod. The lock may include an actuator and a latch movable by the actuator. The latch is movable by the actuator between the locked position vehicle-rearward of the keeper and the unlocked position in which the latch is offset cross-vehicle from the keeper. The actuator may be on the keeper.

[0022] The spring may be on the rod.

[0023] With reference to the Figures, wherein like numerals indicate like parts throughout the several views, a vehicle-bumper assembly **10** includes a bumper **12** and a rod **14** fixed relative to the bumper **12** and elongated along an axis A. A bracket **16** is spaced from the bumper **12** along the axis A of the rod **14**. The rod **14** is movably engaged with the bracket **16** between a design position (FIGS. **5**A-B) and a retracted position (FIG. **5**C). A spring **18** is between the bumper **12** and the bracket **16**. The spring **18** resiliently forces the bumper **12** away from the bracket **16** along the axis A of the rod **14** toward the design position. A keeper **20** is releasably engaged with the bracket **16**. The keeper **20** maintains the rod **14** in the design position when the keeper **20** is engaged with the bracket **16**. A lock **22** is movable from a locked position in which the lock **22** couples the keeper **20** and the bracket **16** and an unlocked position in which the lock **22** is decoupled with the keeper **20** and/or the bracket **16**.

[0024] When the lock **22** is in the locked position, the lock **22** couples the keeper **20** to the bracket **16** to prevent axial movement of the rod **14** vehicle-rearward relative to the bracket **16**. When the lock **22** is in the unlocked position, the lock **22** is decoupled from the keeper **20** and/or the bracket **16** so that the releasable engagement of the keeper **20** with the bracket **16** resists axial movement of the rod **14** vehicle-rearward relative to the bracket **16** unless axial force on the rod **14** reaches a level that releases the keeper **20** from the bracket **16**. In a situation in which axial movement of the rod **14** releases the keeper **20** from the bracket **16**, the bumper **12** and the rod **14** are movable as a unit from the design position toward the retracted position against the force of the spring **18**. As described further below, the lock **22** may be selectively moved between the locked position and the unlocked position, for example, based on vehicle speed. In the event of certain vehicle impacts in which the bumper **12** impacts an object, e.g., a crash test leg form as described further below, the bumper **12** is retractable vehicle rearward from the design position (shown in solid lines in FIG. **1**) to the retracted position (shown in broken lines in FIG. **1**) against the force of the spring **18**. For example, during impact with a leg form, the bumper **12** and rod **14** move vehicle rearward against the force of the spring **18** to absorb energy from the impact and reduce energy delivered from the bumper **12** to the leg form. In such examples, since the spring **18** biases the bumper **12** away from the bracket **16**, when force is decreased or removed from the bumper **12**, the spring **18** may move the bumper **12** toward the design position.

[0025] A vehicle **24** includes the vehicle-bumper assembly **10**. The vehicle **24** includes a vehicle frame **26**. The bracket **16** of the vehicle-bumper assembly **10** is fixed relative to the vehicle frame **26** and spaced from the bumper **12** along the axis A of the rod **14**.

[0026] As described further below, the vehicle-bumper assembly **10** includes an energy-absorber assembly **28** between the vehicle frame **26** and the bumper **12**. The energy-absorber assembly **28** includes the rod **14**, the spring **18**, and the bracket **16**. The energy-absorber assembly **28** supports the bumper **12** on the vehicle frame **26** and selectively allows the bumper **12** to release from the design position to the retracted position in response to detection of certain vehicle impacts. As set forth below, the bumper assembly may include two energy-absorber assemblies that, in combination with each other, support the bumper **12** on the vehicle frame **26**.

[0027] The bumper **12**, as an example, may impact the knee of a pedestrian impact test leg form during a standardized test. The leg form may be a flexible pedestrian leg impactor (Flex-PLI) leg form. Example regulations that can use the leg form include Global Technical Regulation (GTR), ECE R127 and Korean Motor Vehicle **24** Safety Standards (KMVSS). Example new car assessment programs that can use the leg form include EuroNCAP, CNCAP, and ANCAP.

[0028] The vehicle **24** may be any suitable type of automobile, e.g., a passenger or commercial automobile such as a sedan, a coupe, a truck, a sport utility vehicle, a crossover vehicle, a van, a minivan, a taxi, a bus, etc. The vehicle **24**, as an example, may have a relatively high ride height. With reference to FIG. **1**, the vehicle **24** defines a vehicle-longitudinal axis L extending between a front end (not numbered) and a rear end (not numbered) of the vehicle **24**. The vehicle **24** defines a vehicle-lateral axis C extending cross-vehicle from one side to the other side of the vehicle **24**. The vehicle **24** defines a vertical axis V. The vehicle-longitudinal axis L, the vehicle-lateral axis C, and the vertical axis V are perpendicular relative to each other.

[0029] The vehicle **24** includes the vehicle frame **26** and a vehicle body. The vehicle body and the vehicle frame **26** may have a body-on-frame construction (also referred to as a cab-on-frame construction) in which the vehicle body and vehicle frame **26** are separate components, i.e., are modular, and the vehicle body is supported on and affixed to the vehicle frame **26**. In the example shown in the Figures, the vehicle **24** has a body-on-frame construction. As another example, the vehicle body and the vehicle frame **26** may be of a unibody construction in which the vehicle frame **26** is unitary with the vehicle body (including frame rails **30**, pillars, roof rails, etc.). Alternatively, the vehicle frame **26** and vehicle body may have any suitable construction. The vehicle frame **26** and vehicle body may be of any suitable material, for example, steel, aluminum, and/or fiber-reinforced plastic, etc.

[0030] The vehicle body includes body panels (not numbered). The body panels may include structural panels, e.g., rockers, pillars, roof rails, etc. The body panels may include exterior panels. The exterior panels may present a class-A surface, e.g., a finished surface exposed to view by a customer and free of unaesthetic blemishes and defects. The body panels include, e.g., a roof panels, doors, fenders, hood, decklid, etc. The vehicle body may define a passenger cabin to house occupants, if any, of the vehicle **24**.

[0031] The vehicle frame **26** includes frame rails **30** and may include cross beams (not shown). The frame rails **30** are elongated along a vehicle-longitudinal axis L. The frame rails **30** are spaced from each other cross-vehicle. The cross beams of the vehicle frame **26** extend from one frame rail **30** to the other frame rail **30** transverse to the vehicle-longitudinal axis L.

[0032] The vehicle frame **26** includes two frame rails **30**. The frame rails **30** may define the cross-vehicle boundaries of the vehicle frame **26**. The frame rails **30** may be elongated along the vehicle-longitudinal axis L from a rear end of the vehicle **24** to a front end of the vehicle **24**. For example, the frame rails **30** may extend along substantially the entire length of the vehicle **24**. In other examples, the frame rails **30** may be segmented and extend under portions of the vehicle **24**, e.g., at least extending from below a passenger compartment of the vehicle **24** to the front end of the vehicle **24**. In some examples, each frame rail **30** may be unitary from the rear end of the vehicle **24** to the front end of the vehicle **24**. In other examples, the frame rails **30** may include segments fixed to each other (e.g., by welding, threaded fastener, etc.) and in combination extending from a rear end of the vehicle **24** to the front end of the vehicle **24**.

[0033] As set forth above, the vehicle frame **26** may have a body-on-frame construction in which the vehicle body is supported on and affixed to the vehicle frame **26**. In such an example, the frame rails **30** may include cab mount brackets (not shown) on which the vehicle body is supported and affixed. The cab mount brackets are fixed to the frame rails **30**, e.g., welded to the frame rails **30**. The cab mount brackets may extend outboard from the frame rail **30**. The cab mount bracket may be cantilevered from the frame rail **30**. The cab mount brackets are configured to support the vehicle body in a body-on-frame configuration. For example, the cab mount bracket may include a post or a hole that receives a hole or a post, respectively, of the vehicle body to connect the vehicle body to the vehicle frame **26**. Specifically, the vehicle body may be fixed to the cab mount bracket. During assembly of the vehicle **24**, the vehicle body is set on the vehicle frame **26** with fastening features of the vehicle body aligned with the cab mount brackets for engagement with the cab mount brackets.

[0034] The vehicle frame **26** may include suspension and steering attachment points (not shown) that support suspension and steering components of the vehicle **24**. As one example, the suspension and steering attachment points may be suspension towers. Suspension and steering components of the vehicle **24** are connected to the vehicle frame **26**, at least in part, at the suspension towers. The suspension and steering components include suspension shocks, suspension struts, steering arms, steering knuckles, vehicle wheels, etc.

[0035] The vehicle frame **26** may have a powertrain compartment designed (not numbered) to support and house a vehicle powertrain between the frame rails **30**. For example, at least one of the cross-beams of the vehicle frame **26** may be a powertrain cradle, i.e., a cross-beam designed to support and affix to the vehicle powertrain. The powertrain cradle may define a boundary of the powertrain compartment, e.g., a lower boundary of the powertrain compartment. The vehicle powertrain in the powertrain compartment may be, for example, an internal combustion engine and a transmission, in which case the powertrain cradle is an engine cradle. In other examples, the vehicle frame **26**, e.g., the frame rails **30** and/or the cross beams, are designed to support battery assemblies. The battery assembly may be of any suitable type for vehicular electrification to power propulsion of the vehicle **24**, for example, lithium-ion batteries, nickel-metal hydride batteries, lead-acid batteries, or ultracapacitors, as used in, for example, plug-in hybrid electric vehicles (PHEVs), hybrid electric vehicles (HEVs), or battery electric vehicles (BEVs).

[0036] The frame rails **30** and cross-beams may be extruded, roll-formed, etc. The frame rails **30** and cross-beams of the vehicle frame **26** may be of any suitable material, e.g., suitable types of steel, aluminum, and/or fiber-reinforced plastic, etc. The frame rails **30** and cross-beams may be hollow. The frame rails **30** and cross-beams may be rectangular in cross-section (e.g., a hollow rectangular cuboid), round in cross section, e.g., a hollow, round such as a hollow cylinder), etc.

[0037] The vehicle frame **26** includes the frame-rail ends **32** extending vehicle-forward of the frame rails **30**, respectively. In other words, the vehicle frame **26** includes two frame-rail ends **32** with one frame-rail end **32** extending vehicle-forward of one of the frame rails **30** and the other frame-rail end **32** extending vehicle-forward of the other frame rail **30**.

[0038] The frame-rail end **32** is fixed the respective frame rail **30**. For example, the frame-rail end **32** may be fixed to the respective frame rail **30** by welding, fastening, etc. In the example shown in the Figures, the frame-rail end **32** is a component of the vehicle frame **26** that has a body-on-frame architecture, as described above. In other examples, the vehicle frame **26** may be of another architecture, e.g., a unibody architecture. In such examples, the frame rail **30** is a component of the vehicle frame **26** that has a unibody architecture and the frame-rail end **32** is connected to such frame rail **30**.

[0039] The frame-rail end **32** is elongated along the vehicle-longitudinal axis L. For example, the frame-rail end **32** may be coaxial with the frame rail **30** at the connection of the frame-rail end **32** and the frame rail **30**. The frame rail **30** has a vehicle-forward end and the frame-rail end **32** extends vehicle-forward from the vehicle-forward end of the frame rail **30**. Specifically, the frame-rail end **32** has a vehicle-rearward end at the frame rail **30** and a vehicle-forward end, as described further below.

[0040] The frame-rail end **32** includes a base elongated along a vehicle-longitudinal axis L and a flange **34** extending radially from the base. The base may include the vehicle-rearward end of the frame-rail end **32**. The flange **34** may be at the vehicle-forward end of the frame-rail end **32**, as shown in the example shown in the Figures.

[0041] The flange **34** includes a vehicle-forward face. The vehicle-forward face faces in the vehicle-forward direction. The vehicle-forward face may be planar, as shown in the example in the Figures. As set forth below, the bracket **16** abuts the flange **34** at the vehicle-forward end of the frame-rail end **32**. The vehicle-forward face of the flange **34** is at the vehicle-forward end of the frame-rail end **32** in the example shown in the Figures. In other examples, the vehicle-forward face of the flange **34** may be spaced vehicle-rearward of the vehicle-forward end of the frame-rail end **32**.

[0042] The frame-rail end **32** includes mounting holes **36**. Specifically, the mounting holes **36** may be in the flange **34** as shown in the example in the Figures. In such examples, the mounting holes **36** extend through the vehicle-forward face of the flange **34**. The mounting holes **36** mount the bracket **16** to the frame-rail end **32**, as described further below.

[0043] The frame-rail end **32** includes a bore **38** at the vehicle-forward end of the frame-rail end **32**. The bore **38** extends through the vehicle-forward end of the frame-rail end **32**. In other words, the bore **38** is open at the vehicle-forward end of the frame rail **30**. The bore **38** may extend continuously through the frame-rail end **32** through both the vehicle-forward end and the vehicle-rearward end of the frame rail **30**. The bore **38** is elongated along the vehicle-longitudinal axis L. The frame-rail end **32** may be extruded, roll-formed, etc. The frame-rail end **32** may be of any suitable material, e.g., suitable types of steel, aluminum, and/or fiber-reinforced plastic, etc. The frame-rail end **32** may be hollow, i.e., the bore **38** makes the frame-rail end **32** hollow. The frame rails **30** and cross-beams may be rectangular in cross-section (e.g., a hollow rectangular cuboid), round in cross section, e.g., a hollow, round such as a hollow cylinder), etc.

[0044] The frame-rail ends **32** are designed to deform relative to the frame rail **30** during frontal-vehicle impact. Specifically, the frame-rail ends **32** deform vehicle-rearward to allow rearward movement of the vehicle-bumper assembly **10** relative to the frame rails **30** to absorb energy during certain vehicle impacts. The frame-rail ends **32** may include features that direct deformation of the frame-rail end **32** toward the frame rail **30** during frontal impact of the bumper **12**. These features may include wall geometry, wall thickness, dimples, cutouts, etc. The frame-rail ends **32** may be referred to in industry as crush cans.

[0045] With reference to FIGS. **1-3**, the vehicle **24** has a front-end structure. The front-end structure includes a grill and the bumper assembly. The grill is above the bumper assembly. The grill may be a component of the vehicle body and may be supported on other components of the vehicle body.

[0046] The vehicle-bumper assembly **10** includes the bumper **12** and the energy-absorber assembly **28**. The bumper **12** and the energy absorber may be parts-in-assembly (PIA), i.e., assembled as a unit to the vehicle frame **26** at a vehicle assembly plant.

[0047] The vehicle-bumper assembly **10** is connected to the vehicle frame **26**. Specifically, the bumper **12** is connected to the frame-rail ends **32** with the energy-absorber assembly **28**, as described further below.

[0048] The bumper **12** extends transversely to the frame rails **30**, e.g., in a cross-vehicle direction C. With reference to FIGS. **1-3**, the bumper **12** is elongated along the cross-vehicle direction C. The bumper **12** is supported by the vehicle frame **26**, i.e., the weight of the bumper **12** is borne by the vehicle frame **26**. The vehicle-bumper assembly **10** may be a front bumper assembly, as shown in the Figures. In other words, the vehicle-bumper assembly **10** may be at a front of the vehicle **24** and, in such examples, the bumper **12** extension is operable for frontal collisions of the vehicle **24**.

[0049] The vehicle-bumper assembly **10** is supported by the vehicle frame **26**, i.e., the weight of the bumper assembly is borne by the vehicle frame **26**. Specifically, the energy-absorber assembly **28** supports the bumper **12** on the vehicle frame **26**. In other words, the weight of the bumper **12** is borne by the energy-absorber assembly **28** and the weight of the energy-absorber assembly **28** and the bumper **12** is borne by the vehicle frame **26** through the connection of the energy-absorber assembly **28** to the vehicle frame **26**. The vehicle-bumper assembly **10** may be a front bumper assembly, as shown in the example in the Figures. In other words, the bumper assembly may be at a front of the vehicle **24** and, in such examples, the bumper retraction is operable for certain frontal collisions of the vehicle **24**.

[0050] The bumper **12** extends transversely to the frame rails **30**. With reference to FIG. **1**, the bumper **12** is elongated along the vehicle-lateral axis C. The bumper **12** may be of any suitable material such as metal (steel, aluminum, etc.), fiber-reinforced plastic, etc.

[0051] The bumper **12** has a vehicle-forward face and a vehicle-rearward face. The vehicle-forward face may be a class-A surface, i.e., a surface specifically manufactured to have a high-quality, finished aesthetic appearance free of blemishes. As an example, the vehicle-forward face may be chromed. The bumper **12** may be of any suitable material such as metal (steel, aluminum, etc.), fiber-reinforced plastic, etc. The bumper **12** may have a mounting brace on the vehicle-rearward face of the bumper **12**, as shown in the example in the Figures. The mounting brace is fixed to and moves as a unit with the rest of the bumper **12**.

[0052] The bracket **16** is fixed relative to the vehicle frame **26**. In other words, the bracket **16** moves as a unit with the vehicle frame **26**. The bracket **16** may be fixed directly to the vehicle frame **26**. The bracket **16** may be fixed to the vehicle frame **26** by mechanical attachment that requires removal by a service technician with the use of a tool and/or destruction such as cutting, e.g., cutting material and/or welded joints, etc. As an example, as shown in the Figures, the bracket **16** may be fixed directly to the frame-rail end **32**. The bracket **16** may be fixed to the vehicle frame **26** in any suitable way such as fasteners, welding, etc. In the example shown in the Figures, the bracket **16** is fixed directly to the vehicle frame **26** with four threaded studs that extend through the holes in the flange of the frame-rail end **32** with threaded nuts retaining the flange to the bracket **16**.

[0053] The bracket **16** is configured to be fixed relative to a vehicle frame **26**. The bracket **16** is sized and shaped for connection to the vehicle frame **26**. When fixed to the vehicle frame **26**, the bracket **16** is spaced from the bumper **12** vehicle rearward from the bumper **12**. The bracket **16** is spaced from the bumper **12** along the axis A of the rod **14**. The axis A of the rod **14** may be, for example, parallel to the vehicle-longitudinal axis L.

[0054] In the example shown in the Figures, the bracket **16** includes a base **40** and a standoff **42**. The standoff **42** extends vehicle-rearward from the base **40** of the bracket **16**. The standoff **42** may be, for example, cylindrical. The standoff **42** may be annular about the axis A of the rod **14**, as shown in the example in the Figures. In other words, the standoff **42** may be coaxial with the rod **14**. The standoff **42** may define a bore **44** that is coaxial with the rod **14**. The rod **14** may extend into the bore **44** in the design position and the retracted position, as shown in the example in the Figures. The standoff **42** supports the keeper **20**, as described further below. In the example shown in the Figures, the standoff **42**, the keeper **20**, and the lock **22** are in the bore **44** of the frame-rail end **32**, as shown in FIG. **4**.

[0055] The bumper **12** is movable with the rod **14** along the axis A of the rod **14** between the design position and the retracted position during certain vehicle impacts, as described herein. The bracket **16** movably receives the rod **14**. For example, as shown in the example in the Figures, the bracket **16** may define a hole (not numbered) extending through the bracket **16**, i.e., spaced from the outer periphery of the bracket **16** and the hole slidably receives the rod **14**. A retainer **46** is fixed to the rod **14** along the axis A of the rod **14** to retain the rod **14** to the bracket **16** against the force of the spring **18**. In the example shown in the Figures, the rod **14** is threaded and the retainer **46** is threaded, e.g., the retainer **46** is a threaded nut, is threadedly engaged with the rod **14** vehicle rearward of the bracket **16** to retain the rod **14** to the bracket **16**. In such an example, the spring **18** forces the retainer **46** against the bracket **16**. The rod **14** slides along the hole axially along the axis A of the rod **14** and the bumper **12** moves between the design position and the retracted position. The hole may be elongated along the axis A of the rod **14**. In some examples, the hole may be elongated along the vehicle-longitudinal axis L. In other examples, the bracket **16** may include any suitable track, channel, etc., that slidably receives the rod **14**.

[0056] The rod **14** is elongated along the axis A of the rod **14**. In other words, the longest dimension of the rod **14** is along the axis A. The rod **14** may be, for example, cylindrical, as shown in the example in the Figures. The rod **14** may be, for example, metal or any other suitable material. The rod **14**, or the rods **14** in examples including more than one rod **14**, has sufficient rigidity to support the bumper **12** on the vehicle frame **26** and sufficient rigidity to transfer linear movement of the bumper **12** relative to the vehicle frame **26** during movement of the bumper **12** between the design position and the retracted position.

[0057] The rod **14** is fixed relative to the bumper **12** and selectively movable relative to the bracket **16**. The rod **14** may be fixed to the bumper **12** by mechanical attachment that requires removal by a service technician with the use of a tool and/or destruction such as cutting, e.g., cutting material and/or welded joints, etc. In the example shown in the Figures, the rod **14** includes an end extending through the mounting bracket **16** of the bumper **12** that is threaded with opposing threaded fasteners on opposite sides of the mounting bracket **16**.

[0058] The rod **14** is moveable axially relative to the bracket **16** when the lock **22** is in the unlocked position and force on the bumper **12** is sufficient to disengage the retainer from the bracket **16**, e.g., slidable axially through the hole when force applied to the bumper **12** compresses the spring **18** and disengages the retainer from the bracket **16** when the lock **22** is in the disengaged position. The rod **14** is fixed axially relative to the bracket **16** when the lock **22** is in the locked position, i.e., the lock **22** locks the rod **14** to the bracket **16** relative to each other.

[0059] The spring **18** is between the bumper **12** and the bracket **16**. Prior to assembly of the bumper assembly to the vehicle frame **26**, the spring **18** is retained between the bumper **12** and the bracket **16**. Specifically, as described above with respect to the example shown in the Figures, the rod **14** is fixed to the bumper **12** and the retainer **46** retains the rod **14** on the bracket **16** against the force of the spring **18**. The spring **18** resiliently forces the bumper **12** away from the bracket **16** along the axis A of the rod **14** toward the extended position.

[0060] The spring **18** is operatively engaged with the bracket **16** and the bumper **12** to exert force the bumper **12** vehicle-forward away from the bracket **16** and vehicle frame **26** along the axis A. In the example shown in the Figures, the spring **18** abuts the bumper **12** and abuts the bracket **16**. In some examples, the spring **18** is compressed between the bumper **12** and the bracket **16** in the design position. In other examples, the spring **18** may be relaxed in the design position with the spring **18** having a free length equal to the length of the rod **14** extending from the bracket **16** to the bumper **12** in the design position, e.g., when the retainer abuts the bracket **16**. In both examples, the spring **18** maintains the bumper **12** in the design position and prevents vehicle-rearward movement of the bumper **12** relative to the vehicle frame **26** absent forces on the bumper **12** that exceed a force threshold for which the bumper **12** is designed to move toward the retracted position, e.g., during certain vehicle impacts. The force of the spring **18** between the bracket **16** and the bumper **12** is sufficient to prevent rattle of the bumper **12** and vehicle-rearward movement of the bumper **12** during driving of the vehicle **24** absent such forces. The spring **18** is resilient such that, in the event the spring **18** is compressed by vehicle-rearward force on the bumper **12**, the spring **18** decompresses to force the bumper **12** vehicle-forward to the design position when the vehicle-rearward force on the bumper **12** is decreased or removed.

[0061] The spring **18**, as an example, may be a coil spring. In the example shown in the Figures, the spring **18** is a coil spring **18** on the rod **14** between the bumper **12** and the vehicle frame **26**, i.e., the coils of the coil spring **18** helically extend around the rod **14** along the axis A. In such an example, the spring **18** has an inner diameter sized to be received by the rod **14**. The inner diameter of the spring **18** is larger than the outer diameter of the rod **14**. The spring **18** may be metal.
[0062] The keeper **20** maintains the rod **14** in the extended position when the keeper **20** is engaged with the bracket **16**. Specifically, the keeper **20** prevents vehicle-rearward movement of the rod **14** and the bumper **12** relative to the vehicle frame **26** when the keeper **20** is engaged with the bracket **16**.
[0063] The keeper **20** is releasably engaged with the bracket **16**. In other words, the keeper **20** remains engaged with the bracket **16** until a force exceeding a threshold force is applied to the keeper **20** to disengage the keeper **20** from the bracket **16**. The keeper **20** remains engaged with the bracket **16** when force applied to the keeper **20** is below the threshold force.
[0064] The keeper **20** and the bracket **16** are designed so that the keeper **20** releases from the bracket **16** when force above the threshold force is applied to the keeper **20**, specifically, when axial force on the rod **14** from the bumper **12** in the vehicle-rearward direction exceeds the threshold force. The threshold force has a non-zero magnitude. The threshold force may be the vehicle-rearward force on the bumper **12** experienced by the bumper **12** during certain vehicle impacts, e.g., during certain frontal-vehicle impacts with a pedestrian at a vehicle speed of 30 kph-50 kph (kilometers per hour). As an example, the threshold force may be 6 kN. The threshold force may be, for example, empirically determined.
[0065] A releasable connection is between the keeper **20** and bracket **16**. The releasable connection is designed to release the keeper **20** from the bracket **16** when force above the threshold force is applied to the keeper **20**, specifically, when axial force on the rod **14** from the bumper **12** in the vehicle-rearward direction exceeds the threshold force. For example, the releasable connection is designed to release the keeper **20** from the bracket **16** in response to certain frontal-vehicle impacts, as described above.
[0066] In some examples, including the example in FIGS. **3**-**5**C, the releasable connection is a shear pin **48** between the keeper **20** and the bracket **16**, e.g. the standoff **42** of the bracket **16**. The shear pin **48** retains the keeper **20** to the bracket **16** in the absence of application of force above the threshold force on the rod **14** in the vehicle-rearward direction. In the event axial force on the rod **14** from the bumper **12** in the vehicle-rearward direction exceeds the threshold force, the shear pin **48** releases to release the keeper **20** from the bracket **16**. In such examples, the shear pin **48** may undergo plastic shear deformation as the keeper **20** releases from the bracket **16**. The shear pin **48** is designed to release the keeper **20** from the bracket **16** in response to application of axial force on the rod **14** from the bumper **12** in the vehicle-rearward direction exceeds the threshold force. Specifically, the shear pin **48** is sized, shaped, and positioned to release from the keeper **20** and/or the bracket **16**, e.g., by plastic shear deformation of the shear pin **48**, to release the keeper **20** from the bracket **16**.
[0067] The shear pin **48** may be on one of the bracket **16** and the keeper **20** and a hole may be on the other of the bracket **16** and the keeper **20**. The shear pin **48** is received in the hole (not numbered) and is retained in the hole in the absence of application of force above the threshold force. In the event axial force on the rod **14** from the bumper **12** in the vehicle-rearward direction exceeds the threshold force, the shear pin **48** releases from the hole. For example, the shear pin **48** and/or the hole deform to release the shear pin **48** from the hole. In some examples, the shear pin **48** and the hole are designed to release the shear pin **48** from the hole in response to application of axial force on the rod **14** from the bumper **12** in the vehicle-rearward direction exceeds the threshold force. Specifically, the shear pin **48** and hole are sized, shaped, and positioned to release the shear pin **48** from the hole to release the keeper **20** and/or the bracket **16**, e.g., by plastic shear deformation of the shear pin **48**, to release the keeper **20** from the bracket **16**.

[0068] In the example shown in the Figures, the shear pin **48** is on the bracket **16**, specifically the standoff **42** of the bracket **16**, and in other examples the shear pin **48** may be on the keeper **20**. In the example shown in the Figures, the two shear pins **48** are between the keeper **20** and the bracket **16**. In other examples, any suitable number of shear pins **48**, i.e., one or more, may be between the keeper **20** and the bracket **16**.

[0069] In another example, as shown in FIG. **6**, the releasable connection is a keeper spring **50** between the keeper **20** and the bracket **16**. The keeper spring **50** forces the keeper **20** against the standoff **42** in a vehicle-forward direction. The keeper spring **50** is designed (i.e., sized, shaped, positioned, and having a spring **18** constant) to act against forces on the bumper **12** that force the rod **14** to move axial vehicle-rearward against the force of the spring **18**. The keeper spring **50** is designed to retain the keeper **20** to the bracket **16** in the absence of application of force above the threshold force on the rod **14** in the vehicle-rearward direction. In the event axial force on the rod **14** from the bumper **12** in the vehicle-rearward direction exceeds the threshold force, the keeper spring **50** is designed to release the keeper **20** from the bracket **16**. In such examples, the keeper spring **50** undergoes elastic deformation as the keeper **20** releases from the bracket **16** and returns to its original position when the force is released. In other words, the energy-absorber assembly **28** is resettable. The keeper spring **50** may be, for example, a coil spring.

[0070] The bracket **16** may be between the keeper **20** and the bumper **12**. In such an example, the rod **14** extends from the bumper **12** through the bracket **16** and to the keeper **20** in at least the retracted position. In the example shown in the Figures, the rod **14** extends from the bumper **12** through the bracket **16** and to the keeper **20** in both the design position and the retracted position. The rod **14** may abut the keeper **20** in the design position and the retracted position, as shown in the Figures.

[0071] The keeper **20** may be disposed on the axis A of the rod **14**, as shown in the example in the Figures. The keeper **20** is disposed on the axis A of the rod **14** vehicle-rearward of the rod **14**. Specifically, the keeper **20** may extend across the bore **38** with the axis A of the rod **14** projected from the rod **14** is transverse to the to the keeper **20**. In the event of application of force to the bumper **12** that forces the rod **14** to move axially vehicle-rearward against the force of the spring **18**, the rod **14** abuts the keeper **20** and applies a vehicle-rearward force to the keeper **20**. In such an event, if the force on the bumper **12** exceeds the threshold force, the force of the rod **14** on the keeper **20** disengages the keeper **20** from the bracket **16**, e.g., the standoff **42**. In examples including the shear pins **48**, the force of the rod **14** on the keeper **20** plastically deforms the shear pin **48** and/or the hole to release the keeper **20** from the bracket **16**.

[0072] The lock **22** is movable between the locked position and the unlocked position. In the locked position, the lock **22** couples the keeper **20** and the bracket **16**. Specifically, in the locked position, the lock **22** prevents disengagement of the keeper **20** from the bracket **16**. In the unlocked position, the lock **22** is decoupled from the keeper **20** and/or the bracket **16** so that the lock **22** allows for disengagement of the keeper **20** from the bracket **16** in instances when the force on the bumper **12** exceeds the force threshold, as described above. The lock **22** is selectively movable between the locked position and the unlocked position. For example, the lock **22** may be movable between the locked position and the unlocked position based on vehicle speed. In some examples, the lock **22** may move to the locked position at vehicle speeds below 30 kph and above 50 kph and may move to the unlocked position at vehicle speeds between 30 kph and 50 kph.

[0073] The lock **22** is electronically operated based on commands from a vehicle **24** computer, as described further below, e.g., a body control module. The lock **22** may include an actuator **52** and a latch **54** movable by the actuator **52**. The actuator **52** may be a motor, e.g., a DC motor. In the example shown in the Figures, the actuator **52** is a rotary motor that can rotate the actuator **52** in two directions.

[0074] The latch **54** is movable by the actuator **52** between the locked position and the unlocked position. In the locked position, the keeper **20** is between the latch **54** and the bracket **16** and the latch **54** prevents disengagement of the keeper **20** from the bracket **16**. Specifically, the latch **54** prevents vehicle-rearward movement of the keeper **20** relative to the bracket **16**. In the unlocked position, the latch **54** is disengaged with the keeper **20**. When the latch **54** is disengaged with the keeper **20**, the keeper **20** is able to disengage from the bracket **16** in instances when the force on the bumper **12** exceeds the force threshold, as described above. In the example shown in the Figures, in the locked position, the latch **54** is vehicle-rearward of the keeper **20**, and in the unlocked position, the latch **54** is offset cross-vehicle from the keeper **20**. In such examples, the latch **54** being vehicle-rearward of the keeper **20** in the locked position prevents vehicle-rearward movement of the keeper **20**, and the latch **54** being offset cross-vehicle from the keeper **20** in the unlocked position allows for vehicle-rearward movement of the keeper **20** relative to the bracket **16**. The latch **54** in the locked position may abut the keeper **20** or be close to the keeper **20** to a degree such that the keeper **20** cannot disengage from the bracket **16** in the event of force on the bumper **12** that exceeds the force threshold. The latch **54** provides clearance between the latch **54** and the keeper **20** to allow the keeper **20** to move vehicle rearward relative to the bracket **16** when the latch **54** is offset cross-vehicle from the keeper **20**.

[0075] The latch **54** may include an arm **56** that extends from the actuator **52** and a bar **58** that extends from the arm **56**. The actuator **52** moves the arm **56** and the bar **58** relative to the keeper **20** between the locked position and the unlocked position. The bar **58** may be, for example, transverse to the arm **56** at the intersection of the arm **56** and the bar **58**. In other examples, the arm **56** and the bar **58** may curve into each other at the intersection of the arm **56** and the bar **58**. The arm **56** and the bar **58** are fixed relative to each other and move together as a unit. In the example shown in the Figures, the arm **56** and the bar **58** are rotated together as a unit by the actuator **52**. The arm **56** may be releasable from the bar **58**, as described further below.

[0076] The lock **22** may include at least one catch **60** fixed to the bracket **16**. The catch **60** is configured (i.e., sized, shaped, and positioned) to receive the latch **54** in the locked position to retain the latch **54** in the vehicle-rearward position relative to the bracket **16**. In the locked position, the catch **60** anchors the bar **58** to the bracket **16** to prevent vehicle-rearward movement of the bar **58** relative to the bracket **16** and thus prevent vehicle-rearward movement of the keeper **20** relative to the bracket **16**, e.g., in the event of force on the bumper **12** that exceeds the force threshold. In the unlocked position, the bar **58** does not anchor the keeper **20** to the bracket **16**.

[0077] In the example shown in the Figures, two catches **60** are on the bracket **16** on opposite sides of the keeper **20**. When the actuator **52** rotates the locked position, the bar **58** extends across the keeper **20** vehicle-rearward of the keeper **20** from one catch **60** to the other catch **60**. In the example shown in the Figures, then the actuator **52** is in the unlocked position, the bar **58** does not extend vehicle-rearward of the keeper **20**. As an example, in the example shown in the Figures, the bar **58** may remain engaged with one of the catches **60**. In the example shown in the Figures, the arm **56** and the bar **58** rotate together about the axis A as a unit and the arm **56** may be releasable from the bar **58**, as shown in FIG. **6**C. Specifically, one of arm **56** and the bar **58** includes a channel designed to transmit rotational movement and the other of the arm **56** and the bar **58** may be in the channel and may be, in some examples, press fit, adhered, etc., in the channel. In such an example, the connection between the arm **56** and the bar **58** is designed to release in the event of force on the bumper **12** that exceeds the force threshold.

[0078] In the example shown in the Figures, the catch **60** is on the standoff **42** of the bracket **16**. In examples in which the latch **54** includes the arm **56** and the bar **58**, the catches **60** are configured to receive the bar **58** in the locked position. In the example shown in the Figures, the catches **60** have a hole that receive the bar **58**.

[0079] In the example shown in the Figures, the actuator **52** is on the keeper **20**. Specifically, the actuator **52** is supported on the keeper **20**, i.e., the weight of the actuator **52** is borne by the keeper **20**. The is fixed relative to the keeper **20** and moves as a unit with the keeper **20**. In the event the keeper **20** moves relative to the bracket **16** when force on the bumper **12** exceeds the force threshold and the lock **22** is in the unlocked position, the actuator **52** and arms **56** move with the keeper **20** relative to the bracket **16**. In some examples, wiring for the actuator **52** may extend across the keeper **20**.

[0080] In the example shown in the Figures, the arm **56** extends radially from the actuator **52**. The arm **56** is fixed to the actuator **52** and moves as a unit with the actuator **52**. Rotation of the actuator **52**, e.g., rotation of a shaft of the motor, rotates the arms **56** about the actuator **52** to extend the bar **58** across the catches **60**. When the bar **58** extends across the catches **60**, the connections of the bar **58** to the catches **60** anchor the keeper **20** to the bracket **16**. In the example shown in the Figures, two catches **60** are on the bracket **16** and two latches **54** extend radially from the actuator **52** to engage the respective catch **60**. Other examples include any suitable number of pairs of catches **60** and latches **54**, i.e., one or more.

[0081] With reference to FIG. **7**, the vehicle **24** includes the vehicle computer **62** a processor and a memory. The computer **62** may be a body control module. The memory includes one or more forms of computer readable media, and stores instructions executable by the computer for performing various operations, including as disclosed herein and including, for example, process shown in FIG. **8** and described below. For example, the computer **62** may be a generic computer with a processor and memory as described above and/or may include an electronic control unit ECU or controller for a specific function or set of functions, and/or a dedicated electronic circuit including an ASIC (application specific integrated circuit) that is manufactured for a particular operation, e.g., an ASIC for processing sensor data and/or communicating the sensor data. In another example, the computer **62** may include an FPGA (Field-Programmable Gate Array) which is an integrated circuit manufactured to be configurable by a user. Typically, a hardware description language such as VHDL (Very High-Speed Integrated Circuit Hardware Description Language) is used in electronic design automation to describe digital and mixed-signal systems such as FPGA and ASIC. For example, an ASIC is manufactured based on VHDL programming provided pre-manufacturing, whereas logical components inside an FPGA may be configured based on VHDL programming, e.g., stored in a memory electrically connected to the FPGA circuit. In some examples, a combination of processor(s), ASIC(s), and/or FPGA circuits may be included in the computer. The memory may be of any type, e.g., hard disk drives, solid state drives, servers, or any volatile or non-volatile media. The memory may store the collected data sent from the sensors. The memory may be a separate device from the computer **62**, and the computer may retrieve information stored by the memory via a vehicle communication network **64**, e.g., over a CAN bus, a wireless network, etc. Alternatively or additionally, the memory may be part of the computer, e.g., as a memory of the computer **62**.

[0082] As shown in FIG. **7**, the computer **62** is generally arranged for communications on the vehicle communication network **64** that may include a bus in the vehicle **24** such as a controller area network CAN or the like, and/or other wired and/or wireless mechanisms. Alternatively or additionally, in cases where the computer **62** includes a plurality of devices, the vehicle communication network **64** may be used for communications between devices represented as the computer **62** in this disclosure. Further, as mentioned below, various controllers and/or sensors may provide data to the computer **62** via the vehicle communication network **64**.

[0083] With reference to FIG. **7**, the vehicle **24** may include at least one impact sensor **66** for sensing certain vehicle impacts (e.g., impacts of a certain magnitude, direction, etc.) during and/or prior to impact. “Certain” indicates the type and/or magnitude of the impact. The type and/or magnitude of such “certain vehicle impacts” may be pre-stored in the computer, e.g., a restraints control module and/or a body control module. The impact sensor **66** may be of any suitable type, for example, post contact sensors such as accelerometers, pressure sensors, and contact switches; and pre-impact sensors **66** such as radar, LIDAR, and vision sensing systems. The vision sensing systems may include one or more cameras, CCD image sensors, CMOS image sensors, etc. The impact sensor **66** may be located at numerous points in or on the vehicle **24**. The impact sensor **66** may be in communication with the computer **62**.

[0084] With reference to FIG. **7**, the vehicle **24** includes a speed sensor **68** that detects the speed the vehicle **24** is moving relative to ground. The speed sensor **68** may be of any suitable type, including, in some examples, those known in the art. The speed sensor **68** may be, for example, a wheel-speed sensors **68** that measuring the rotational speed of a vehicle wheel. In other examples, the speed sensor **68** may be of any suitable type. The speed sensor **68** is in communication with the vehicle computer **62**, e.g., through the vehicle communication network **64**.

[0085] With reference to FIG. **8**, the computer **62** stores instructions to control components of the vehicle **24** according to the method **800** shown in FIG. **8**. Use of “in response to,” “based on,” and “upon determining” herein, including with reference to FIG. **7**, indicates a causal relationship, not merely a temporal relationship.

[0086] As shown in FIG. **8**, the method includes moving the lock **22** to the unlocked position when the speed of the vehicle **24** is within a predetermined speed range and move the lock **22** to the locked position when the speed of the vehicle **24** is outside of the predetermined speed range. As an example, the predetermined speed range may have a lower boundary above zero. In such examples, the speed of the vehicle **24** may be outside of the predetermined speed range by being above or below the predetermined speed range. As one example, the predetermined speed range may be 30 kph-50 kph. The predetermined speed range is stored in the vehicle **24** computer, i.e., is predetermined.

[0087] With reference to block **805**, the method includes determining whether the speed of the vehicle **24** is within the predetermined speed range. The vehicle **24** computer may determine whether the speed of the vehicle **24** is within the predetermined speed range based on data from the speed sensor **68** indicating the speed that the vehicle **24** is traveling.

[0088] With reference to block **810**, if the speed of the vehicle **24** is outside of the predetermined speed range, the method **800** includes commanding the actuator **52** to move the lock **22** to the locked position. For example, in the example shown in the Figures, the actuator **52** rotates the arm **56** to engage the bar **58** with the catch **60**. The actuator **52** is maintained in this position while the vehicle **24** travels at a speed outside of the predetermined speed range.

[0089] With reference to block **815**, if the speed of the vehicle **24** is within the predetermined speed range, the method **800** includes moving the actuator **52** to the disengaged position. For example, in the example shown in the Figures, the actuator **52** rotates the arm **56** to disengage the bar **58** from the catch **60**. The actuator **52** is maintained in this position while the vehicle **24** travels at a speed inside of the predetermined speed range.

[0090] The disclosure has been described in an illustrative manner, and it is to be understood that the terminology which has been used is intended to be in the nature of words of description rather than of limitation. Many modifications and variations of the present disclosure are possible in light of the above teachings, and the disclosure may be practiced otherwise than as specifically described.

Claims

1. A vehicle-bumper assembly comprising: a bumper; a rod fixed relative to the bumper and elongated along an axis; a bracket spaced from the bumper along the axis of the rod, the rod being movably engaged with the bracket between a design position and a retracted position; a spring between the bumper and the bracket, the spring resiliently forcing the bumper away from the bracket along the axis of the rod toward the design position; a keeper releasably engaged with the bracket, the keeper maintaining the rod in the design position when the keeper is engaged with the bracket; and a lock movable between a locked position in which the lock couples the keeper and the bracket and an unlocked position in which the lock is decoupled from the keeper and/or the bracket.

2. The vehicle-bumper assembly of claim 1, further comprising a releasable connection between the keeper and bracket.
3. The vehicle-bumper assembly as set forth in claim 2, wherein the releasable connection is designed to release the keeper from the bracket in response to certain frontal-vehicle impacts.
4. The vehicle-bumper assembly as set forth in claim 1, further comprising a shear pin between the keeper and the bracket.
5. The vehicle-bumper assembly as set forth in claim 1, wherein the rod extends through the bracket and the bracket is between the keeper and the bumper.
6. The vehicle-bumper assembly as set forth in claim 5, wherein the keeper is disposed on the axis of the rod.
7. The vehicle-bumper assembly as set forth in claim 6, wherein the lock includes an actuator and a latch movable by the actuator.
8. The vehicle-bumper assembly as set forth in claim 7, wherein the latch is movable by the actuator between the locked position in which the keeper is between the latch and the bracket and the unlocked position in which the latch is disengaged with the keeper.
9. The vehicle-bumper assembly as set forth in claim 7, wherein the actuator is on the keeper.
10. The vehicle-bumper assembly as set forth in claim 1, wherein the spring is on the rod.
11. A vehicle comprising: a vehicle frame; a bumper; a rod fixed relative to the bumper and elongated along an axis; a bracket fixed relative to the vehicle frame and spaced from the bumper along the axis of the rod, the rod being movably engaged with the bracket between a design position and a retracted position; a spring between the bumper and the bracket, the spring resiliently forcing the bumper away from the bracket along the axis of the rod toward the design position; a keeper releasably engaged with the bracket, the keeper maintaining the rod in the design position when the keeper is engaged with the bracket; and a lock movable between a locked position in which the lock couples the keeper and the bracket and an unlocked position in which the lock decouples the keeper and/or the bracket.
12. The vehicle-bumper assembly of claim 11, further comprising a releasable connection between the keeper and bracket.
13. The vehicle-bumper assembly as set forth in claim 12, wherein the releasable connection is designed to release the keeper from the bracket in response to certain frontal-vehicle impacts.
14. The vehicle-bumper assembly as set forth in claim 11, further comprising a shear pin between the keeper and the bracket.
15. The vehicle-bumper assembly as set forth in claim 11, wherein the rod extends through the bracket and the keeper is on a vehicle-rearward side of the bracket.
16. The vehicle-bumper assembly as set forth in claim 15, wherein the keeper is disposed on the axis of the rod vehicle-rearward of the rod.
17. The vehicle-bumper assembly as set forth in claim 16, wherein the lock includes an actuator and a latch movable by the actuator.
18. The vehicle-bumper assembly as set forth in claim 17, wherein the latch is movable by the actuator between the locked position vehicle-rearward of the keeper and the unlocked position in which the latch is offset cross-vehicle from the keeper.
19. The vehicle-bumper assembly as set forth in claim 17, wherein the actuator is on the keeper.
20. The vehicle-bumper assembly as set forth in claim 11, wherein the spring is on the rod.